

**SIMATS ENGINEERING
ASSESSMENT TOOLS****COURSE NAME:** Theory of Computation**COURSE CODE:** CSA13

COURSE OUTCOME COVERED**CO5: Design Pushdown Automata and analyze their equivalence with Context-Free Grammars through formal construction and proof techniques. (BL :3)****Assessment Tool Weightage: 40%**

QUESTIONS: 5
Duration : 1 Hour**Total Marks:25****Constraint Based Problem**

Code of Conduct:

I V. Smylika (192342115) (Name / Reg No) certify that this submission is my original work and adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Smylika V
Signature of the student:

Given: $L = \{a^n b^n \mid n \geq 1\}$

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CSAIBot

Constraint

Only one stack symbol is allowed besides the initial stack symbol.

Let:

- Input alphabet: $\Sigma = \{a, b\}$
- Stack alphabet: $\Gamma = \{z_0, x\}$
- z_0 = Initial stack symbol.
- x = Only additional stack symbol.

PPDA states

- q_0 : Read a's and push x
- q_1 : Read b's and pop x
- q_f : Final state.

$$M = \{Q, \Sigma, \Gamma, \delta, q_0, z_0, F\}$$

where

$$Q = \{q_0, q_1, q_f\}$$

$$\Sigma = \{a, b\}$$

$$\Gamma = \{z_0, x\}$$

$$F = \{q_f\}$$

Transition Function:

$$\delta(q_0, a, z_0) = \{(q_0, xz_0)\}$$

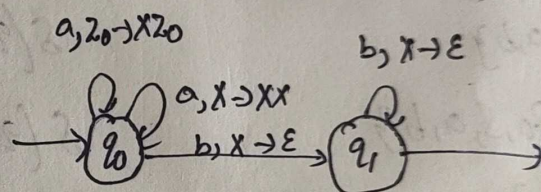
$$\delta(q_0, a, x) = \{(q_0, xx)\}$$

$$\delta(q_0, b, x) = \{(q_1, \epsilon)\}$$

$$\delta(q_1, b, x) = \{(q_1, \epsilon)\}$$

$$\delta(q_1, \epsilon, z_0) = \{(q_f, z_0)\}$$

State Diagram



z_0 = Initial stack symbol

x = Only additional stack

ϵ = empty string

Ex: $aaabbb$: Input stack: $a a a b b b$

z_0

xz_0

$xxz_0, \quad xxxz_0, \quad xxxz_0, \quad xxxz_0$

$\therefore aaabbb$ is accepted.

Result: The PDA accepts exactly $L = \{a^n b^n \mid n \geq 1\}$ using only one

2. CFG to PDA conversion with limited productions. Given a Context-free

$$S \rightarrow aSb \mid ab.$$

CFG to PDA conversion

Given CFG:

$$S \rightarrow aSb \mid ab$$

Constraint

The PDA must have only 2 states.

We use the standard CFG-to-PDA construction with:

- q_0 : start state
- q_1 : processing / accept state.

The PDA uses the grammar variable S as a stack symbol.

PDA

$$M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$$

$$\text{where } Q = \{q_0, q_1\}$$

$$\Sigma = \{a, b\}$$

$$\Gamma = \{z_0, S, a, b\}$$

$$F = \{q_1\}$$

Transitions

$$\delta(q_0, \epsilon, z_0) = \{(q_1, Sz_0)\}$$

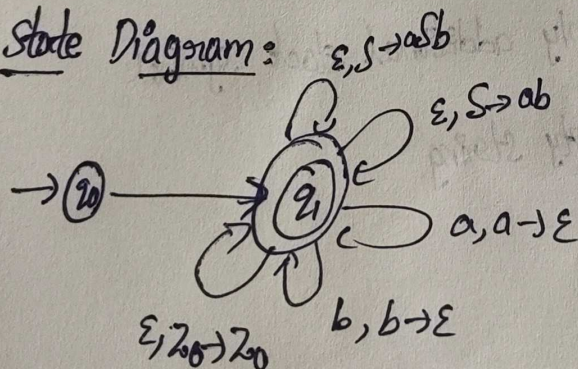
$$\delta(q_1, \epsilon, S) = \{(q_1, ab)\}$$

$$\delta(q_1, a, a) = \{(q_1, \epsilon)\}$$

$$\delta(q_1, b, b) = \{(q_1, \epsilon)\}$$

$$\delta(q_1, \epsilon, z_0) = \{(q_1, z_0)\}$$

State Diagram:



ϵ = empty string

z_0 = Initial stack symbol

S = start

Ex: for $n=3$:

$$S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aaabbbb$$

Result: The equivalent PDA uses exactly 2 states.

to CFG conversion with at Most 3 variables
since the actual PDA to be converted is not specified in the question, we can use a standard PDA for $L = \{a^n b^n | n \geq 1\}$ and construct an equivalent CFG.

The PDA has:

• q_0 : pushing state

• q_1 : popping state

• q_f : accepting state.

A conventional PDA-to-CFG conversion can create many variables of the
However, the question restricts the CFG to at most 3 variables.
we can use an equivalent simplified CFG:

$$S \rightarrow aSb / ab$$

This has only one non-terminal.

Therefore, it satisfies the maximum of 3 variables.

Derivation:

For $n=1$:

$$S \Rightarrow ab$$

For $n=2$:

$$S \Rightarrow aSb \Rightarrow aaSbb$$

For $n=3$:

$$S \Rightarrow aSb \Rightarrow aaSbbb \Rightarrow aaabbbb$$

Thus,

$$L(G) = \{a^n b^n | n \geq 1\}$$

Result:

equivalent CFG:

$$S \rightarrow aSb / ab$$

Number of variables:

$$1 \leq 3$$

4. Turing Machine For $L = \{ww \mid w \in \{0,1\}^*$

The stated constraint is:

The tape head can move only to the right; no left movement is allowed.

A machine whose head can only move right processes the input in a one-way streaming fashion.

Once the machine moves past an input symbol,

it can never return to that symbol.

To recognize $L = \{ww\}$

The machine needs to compare the first half of the string with the second half.

For example:

$w = 01$

Then: $ww = 0101$

The machine must compare:

A one-way machine cannot go back to the first 01 to perform arbitrary comparisons.

In fact, a one-way ~~TM~~ with no left movement and finite cannot recognize arbitrary nonregular languages such as ww^* ; it behaves like a finite-state streaming machine with respect to the unread input.
 $\therefore$ No standard Turing machine satisfying.

But this requires left movement, so it violates the given

SIMATS ENGINEERING ASSESSMENT TOOLS

CRITERIA FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

94/100